

Spot the Unfair Test — Set B — Answer Key

Level 4 · Ages 10-13 · Answers in bold. Accept any answer that means the same.

- ① Aisha wants to find out whether sugar or salt dissolves faster. She stirs a spoonful of sugar into hot water and a spoonful of salt into cold water, and times each one.

What is the test trying to find out?

Whether the type of solid (sugar or salt) affects how fast it dissolves.

What is wrong with it? **The water temperature changed too, so she cannot tell which one made the difference. (Also accept: tested once.)**

How would you fix it? **Use the same volume of water at the same temperature, the same mass of each solid, and the same stirring.**

- ② Leo wants to know if heavier toy cars roll farther. He rolls a light car and a heavy car down the same ramp once each and measures how far each one goes.

What is the test trying to find out?

Whether the mass of a car affects how far it rolls.

What is wrong with it? **Each car was tested only once, so one odd result could mislead. The cars may also differ in more than mass (wheels, shape).**

How would you fix it? **Repeat each run several times and use the mean. Better: use one car and add masses to it.**

- ③ A class gives a new plant food to 10 bean plants. After three weeks the plants have grown 12 cm on average, so the class decides that the plant food works.

What is the test trying to find out?

Whether the plant food makes bean plants grow taller.

What is wrong with it? **There is no control group. Without plants that got no plant food, they cannot tell if the plants would have grown 12 cm anyway.**

How would you fix it? **Grow an identical group of 10 plants with no plant food in the same conditions, and compare the two groups.**
